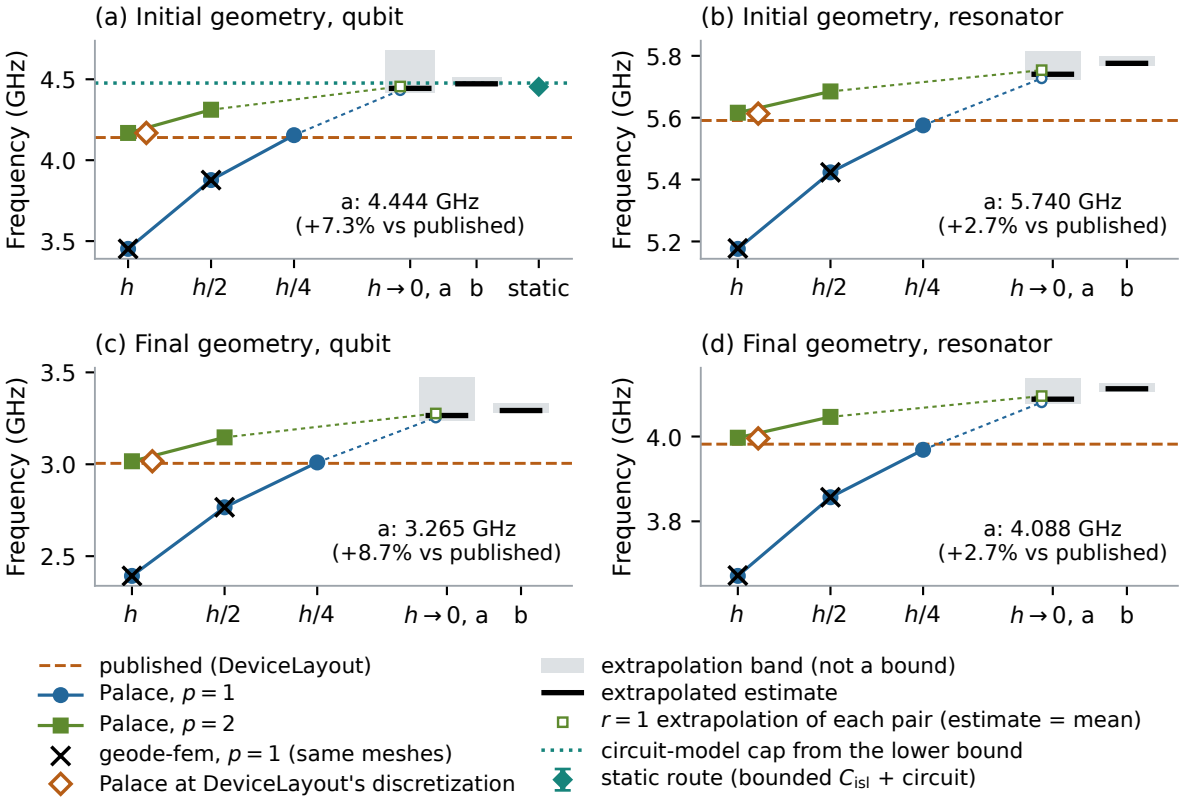
(a) Initial geometry, qubit
Frequency (GHz)
4.5
4.0
3.5
a: 4.444 GHz
(+7.3% vs published)
h
h/2
h/4
h → 0, a
b
static
(b) Initial geometry, resonator
Frequency (GHz)
5.8
5.6
5.4
5.2
a: 5.740 GHz
(+2.7% vs published)
h
h/2
h/4
h → 0, a
b
(c) Final geometry, qubit
Frequency (GHz)
3.5
3.0
2.5
a: 3.265 GHz
(+8.7% vs published)
h
h/2
h/4
h → 0, a
b
(d) Final geometry, resonator
Frequency (GHz)
4.0
3.8
a: 4.088 GHz
(+2.7% vs published)
h
h/2
h/4
h → 0, a
b
published (DeviceLayout)
Palace, p = 1
Palace, p = 2
geode-fem, p = 1 (same meshes)
Palace at DeviceLayout's discretization
extrapolation band (not a bound)
extrapolated estimate
r = 1 extrapolation of each pair (estimate = mean)
circuit-model cap from the lower bound
static route (bounded C_isl + circuit)
h → 0 slots (categorical): a = gmsh split of h; b = Palace refinement of h; static = bounded C_isl + circuit